4 Two Port Networks

INTRODUCTION

Network having two port, one for applying the input and the other for connecting the load is said to be two port network.

When we model a network in terms of two port parameters, we are actually interested in the parameters of the two terminals rather than how does a the network operate inside.

Two port parameters govern the relationship between voltages and currents at the two ends.

One port network:

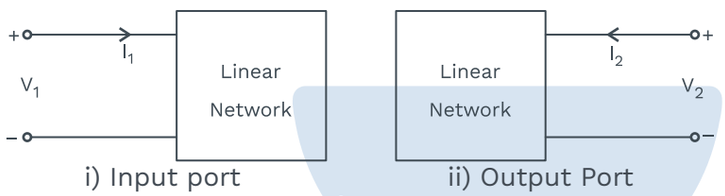


Fig. 4.1

Two port network:

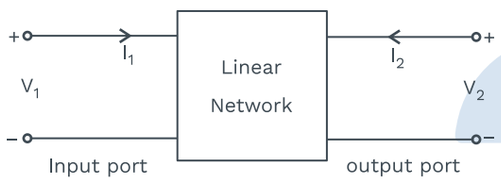


Fig. 4.2 Representation of Two Port Network

Definitions

Port: It has two terminals in which the current enters at one terminal and leaves at another terminal.

V_1 and I_1 → Input port variables

V_2 and I_2 → Output port variables

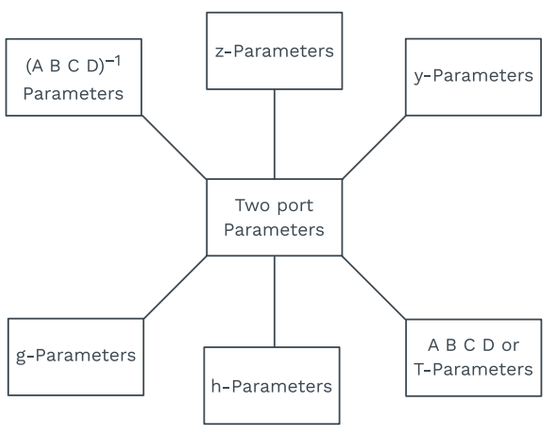


Fig. 4.3 Representation of Types of the Two Port Parameters

- Here, V_1 and V_2 are not the supply or output voltages. These are input and output voltages, respectively.
- Similarly, I_1 and I_2 are not the supply or output currents. These are input and output currents, respectively.

Z-parameters:

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$
$$V_2 = Z_{21} I_1 + Z_{22} I_2$$

Here, $V_1 = f(I_1, I_2)$

$V_2 = f(I_1, I_2)$

- V_1 and V_2 are dependent variables.
- I_1 and I_2 are independent variables.

Grey Matter Alert!

If $I_1 = f(I_2)$ or $I_2 = f(I_1)$, then Z–parameter does not exist.

Matrix form:

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

$Z_{11} = \left.\frac{V_1}{I_1}\right|_{I_2=0}$ = driving point input impedance, when I_2 = 0A.

$Z_{21} = \left.\frac{V_2}{I_1}\right|_{I_2=0}$ = output transfer impedance, when I_2 = 0A.

$Z_{22} = \left.\frac{V_2}{I_2}\right|_{I_1=0}$ = driving point output impedance, when I_1 = 0A.

$Z_{12} = \left.\frac{V_1}{I_2}\right|_{I_1=0}$ = input transfer impedance, when I_1 = 0A.

- Z-parameters are also known as open circuit parameters.
- Impedances of Z-matrix are known as conditional impedances.

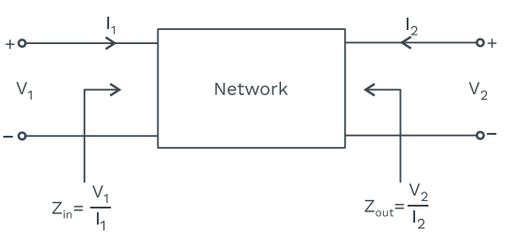


Fig. 4.4 Representation of Network with Z Parameters

$Z_{11} = \left.\frac{V_1}{I_1}\right|_{I_2=0} \neq Z_{in}$

$$Z_{22} = \left.\frac{V_2}{I_2}\right|_{I_1=0} \neq Z_{out}$$

Here,
Z_{11} and Z_{22} are conditional impedances.
Z_{in} and Z_{out} are unconditional impedances.

Grey Matter Alert!

If the ratio of response (output) to the excitation (input) is constant, then that network is reciprocal.

Modelling of [Z] -parameters:

$V_1 = Z_{11} I_1 + Z_{12} I_2$
$V_2 = Z_{21} I_1 + Z_{22} I_2$
$[Z_{11} = Z_{22}] \rightarrow$ Symmetrical network
$[Z_{12} = Z_{21}] \rightarrow$ Reciprocal network

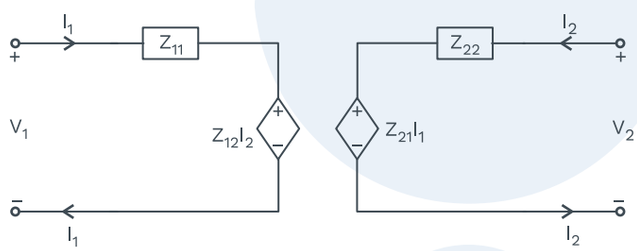


Fig. 4.5 Network Modeled with Z Parameter

Case I (T-network):

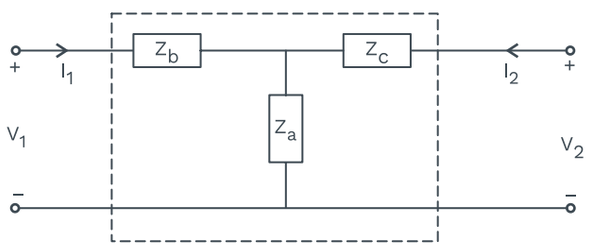


Fig. 4.6 Representation of T - Network

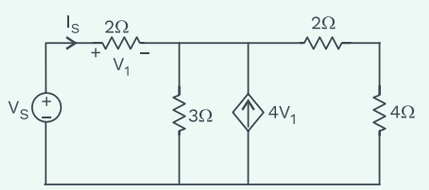
T-network

$$[Z] = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix}$$

$$\begin{Bmatrix} V_1 = Z_{11} I_1 + Z_{12} I_2 \\ V_2 = Z_{21} I_1 + Z_{22} I_2 \end{Bmatrix}$$

Previous Years' Question

The driving point input impedance seen from the source V_S of the circuit shown below, in Ω, is_____.



i) When I_2 = 0A [port-02 is open circuited]

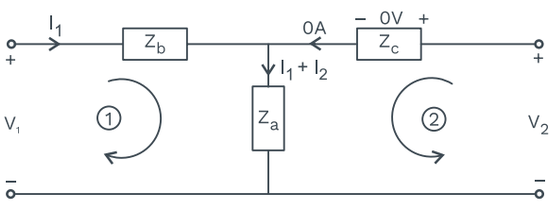


Fig. 4.7 Representing Current Distribution in Two Port Network

Applying KVL in loop 1),

$V_1 - I_1 (Z_b) - I_1 (Z_a) = 0$

$V_1 = (Z_a + Z_b) I_1$

$$\left.\frac{V_1}{I_1}\right|_{I_2=0} = Z_a + Z_b = Z_{11}$$

Applying KVL in loop 2),

$V_2 = I_1 Z_a$

$$\left.\frac{V_2}{I_1}\right|_{I_2=0} = Z_a = Z_{21}$$

ii) When I_1 = 0A [port-1 is open circuited]

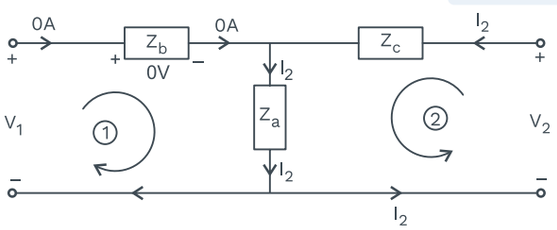


Fig. 4.8

Apply KVL in loop 1),

$V_1 = I_2 Z_a$

$$\left.\frac{V_1}{I_2}\right|_{I_1=0} = Z_a = Z_{12}$$

Apply KVL in loop 2),

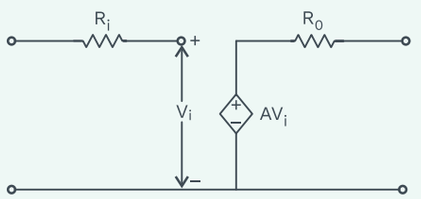
$V_2 - I_2 Z_c - I_2 Z_a = 0$

$V_2 = I_2(Z_a + Z_c)$

$$\left.\frac{V_2}{I_2}\right|_{I_1=0} = Z_a + Z_c = Z_{22}$$

Previous Years' Question

The parameters of the circuit shown in the figure are R_i = 1MΩ, R_0 = 10Ω, A = 10^6V/V. If V_i = 1 μV, the output voltage, input impedance and output impedance respectively are:



a) 1 V, ∞, 10Ω
b) 1V, 0, 10Ω
c) 1V, 0, ∞
d) 10V, ∞, 10Ω

$$[Z] = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} Z_a + Z_b & Z_a \\ Z_a & Z_a + Z_c \end{bmatrix}$$

Case II [lattice network]:

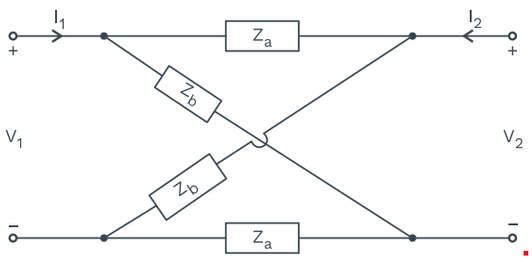


Fig. 4.9

$Z_a \rightarrow$ Series arm impedance

$Z_b \rightarrow$ Diagonal impedance.

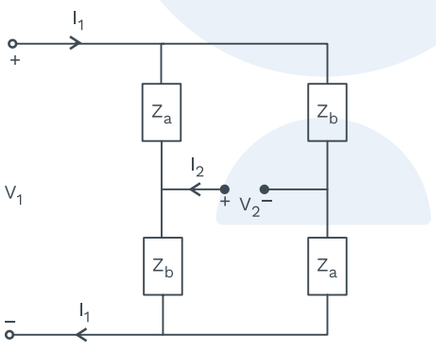


Fig. 4.10

Case I [when I_2 = 0A]

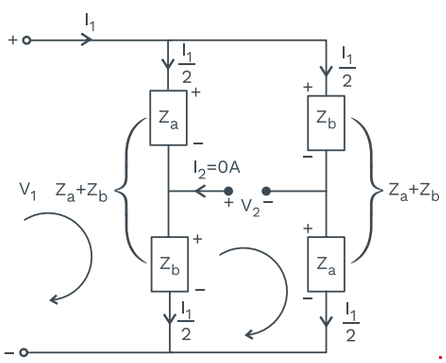


Fig. 4.11

$$V_1 = (Z_a + Z_b)\frac{I_1}{2}$$

$$\left.\frac{V_1}{I_1}\right|_{I_2=0} = \frac{Z_a + Z_b}{2} = Z_{11}$$

$$-V_2 - Z_a \times \frac{I_1}{2} + Z_b \frac{I_1}{2} = 0$$

$$V_2 = I_1\left[\frac{Z_b - Z_a}{2}\right]$$

$$\left.\frac{V_2}{I_1}\right|_{I_2=0} = \frac{Z_b - Z_a}{2} = Z_{21}$$

Case 02 [when $I_1 = 0A$]

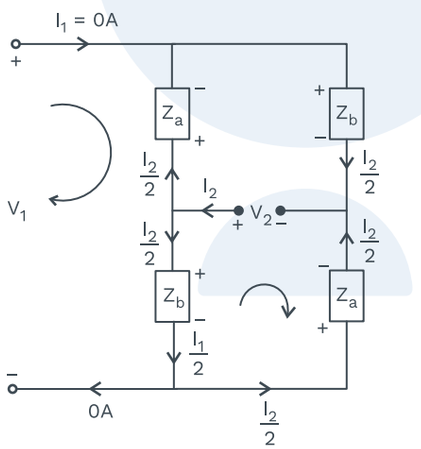


Fig. 4.12

$$V_1 + \frac{I_2}{2}Z_a - \frac{I_2}{2}Z_b = 0$$

$$V_1 = \left(\frac{Z_b - Z_a}{2}\right)I_2$$

$$\left.\frac{V_1}{I_2}\right|_{I_1=0} = \left(\frac{Z_b - Z_a}{2}\right) = Z_{12}$$

$$-V_2 + \frac{I_2}{2}Z_a + \frac{I_2}{2}Z_b = 0$$

$$V_2 = \left[\frac{Z_a + Z_b}{2}\right]I_2$$

Rack Your Brain

What is the condition of symmetrical and reciprocal in lattice network?

Grey Matter Alert!

For the existence of [Y]-parameters:
$V_1 \neq f(V_2)$ or $V_2 \neq f(V_1)$

$$\left.\frac{V_2}{I_2}\right|_{I_1=0} = \left[\frac{Z_a + Z_b}{2}\right] = Z_{22}$$

$$[Z]_{lattice} = \begin{bmatrix} \left(\frac{Z_a + Z_b}{2}\right) & \left(\frac{Z_b - Z_a}{2}\right) \\ \left(\frac{Z_b - Z_a}{2}\right) & \left(\frac{Z_a + Z_b}{2}\right) \end{bmatrix}$$

Y-parameters:

$I_1 = f(V_1, V_2)$

$I_2 = f(V_1, V_2)$

I_1 and I_2 are dependent variables.

V_1 and V_2 are independent variables.

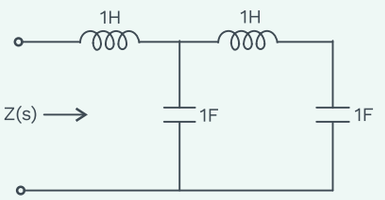
$I_1 = Y_{11}V_1 + Y_{12}V_2$

$I_2 = Y_{21}V_1 + Y_{22}V_2$

$$[Y] = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix}$$

Previous Years' Question

The driving point impedance Z(s) for the circuit shown below is:



a) $\frac{s^4 + 3s^2 + 1}{s^3 + 2s}$ **b)** $\frac{s^4 + 2s^2 + 4}{s^2 + 2}$

c) $\frac{s^2 + 1}{s^4 + s^2 + 1}$ **d)** $\frac{s^3 + 1}{s^4 + s^2 + 1}$

Modelling of Y-parameters:

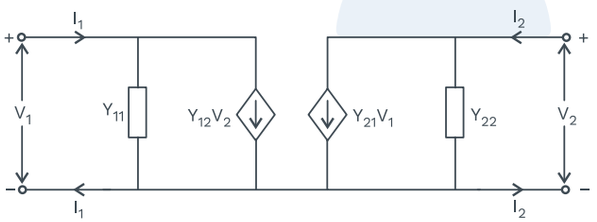


Fig. 4.13 Representation of Modeling of Y Parameters

Case I when V_2 = 0V [port-2 is short circuited]

$$Y_{11} = \left.\frac{I_1}{V_1}\right|_{V_2=0} = \text{Input driving point admittance}$$

$$Y_{21} = \left.\frac{I_2}{V_1}\right|_{V_2=0} = \text{Output transfer admittance}$$

Case II when V_1 = 0V [port-1 is short circuited]

$$Y_{22} = \left.\frac{I_2}{V_2}\right|_{V_1=0} = \text{Output driving point admittance}$$

$$Y_{12} = \left.\frac{I_1}{V_2}\right|_{V_1=0}$$ = Input transfer admittance

- Y-parameters are also known as "short-circuit" parameters as well as conditional parameters.

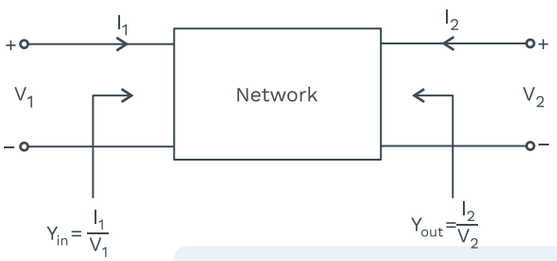


Fig. 4.14

Y_{in} = Input admittance = $\left(\frac{I_1}{V_1}\right) \neq Y_{11}$

Here,

Y_{in} is unconditional input admittance

Y_{out} = Output admittance = $\left(\frac{I_2}{V_2}\right) \neq Y_{22}$

Y_{out} = Unconditional output admittance

- [Y] -parameters in terms of [Z] parameters and vice versa.

$$Y_{11} \neq \frac{1}{Z_{11}},\ Y_{22} \neq \frac{1}{Z_{22}},\ Y_{21} \neq \frac{1}{Z_{21}},\ Y_{12} \neq \frac{1}{Z_{12}}$$

Because both are calculated using different conditions.

$$[Y] = [Z]^{-1} = \frac{\text{Adj}[Z]}{\det[Z]}$$

or

$$[Z] = [Y]^{-1} = \frac{\text{Adj}[Y]}{\det[Y]}$$

Rack Your Brain

What is the relation between [Z] and [Y] parameters?

Note:

This condition is not always true. It depends on the existence of parameters.

π – network

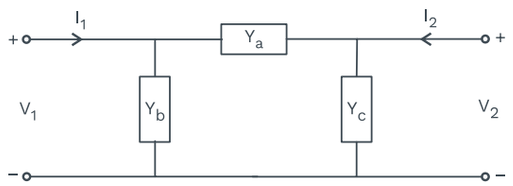


Fig. 4.15

$$Y_a = \frac{1}{Z_a}, Y_b = \frac{1}{Z_b}, Y_c = \frac{1}{Z_c}$$

$$\left.\begin{matrix} I_1 = Y_{11}V_1 + Y_{12}V_2 \\ I_2 = Y_{21}V_1 + Y_{22}V_2 \end{matrix}\right\}$$

Case I [V_2 = 0V]

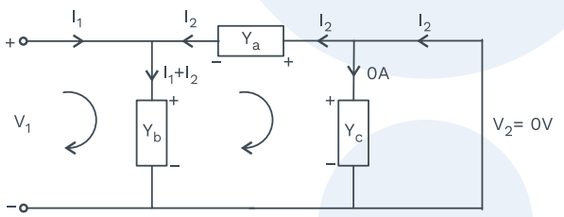


Fig. 4.16

$$V_1 = \frac{(I_1 + I_2)}{Y_b} \quad \ldots (1)$$

$$\frac{(I_1 + I_2)}{Y_b} + \frac{I_2}{Y_a} = 0 \quad \ldots (2)$$

$$I_2\left[\frac{1}{Y_b} + \frac{1}{Y_a}\right] = \frac{-I_1}{Y_b}$$

$$I_2 = \frac{-I_1 Y_a}{Y_a + Y_b}$$

$$V_1 = \frac{I_1}{Y_b} - \frac{I_1 Y_a}{Y_b(Y_a + Y_b)}$$

$$V_1 = \frac{I_1}{Y_b}\left[1 - \frac{Y_a}{Y_a + Y_b}\right]$$

$$\frac{V_1}{I_1} = \frac{1}{Y_b}\left[\frac{Y_a + Y_b - Y_a}{Y_a + Y_b}\right]$$

$$\frac{V_1}{I_1} = \frac{1}{Y_a + Y_b}$$

$$Y_{11} = \left.\frac{I_1}{V_1}\right|_{V_2=0} = Y_a + Y_b = Y_{11}$$

$$I_1 = -I_2\left[\frac{Y_a + Y_b}{Y_a}\right]$$

$$V_1 = -\frac{I_2(Y_a + Y_b)}{Y_b Y_a} + \frac{I_2}{Y_b}$$

$$V_1 = \frac{I_2}{Y_b}\left[1 - \frac{Y_a + Y_b}{Y_a}\right]$$

$$V_1 = \frac{I_2}{Y_b}\left[\frac{Y_a - Y_a - Y_b}{Y_a}\right]$$

$$\frac{I_2}{V_1} = -Y_a$$

$$\left.\frac{I_2}{V_1}\right|_{V_2=0} = -Y_a = Y_{21}$$

Case II [V_1 = 0V]

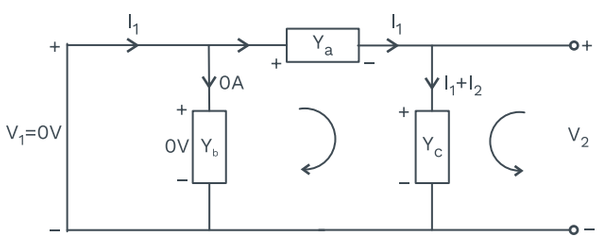


Fig. 4.17

$$V_2 = \frac{I_1 + I_2}{Y_c} \quad \text{... (1)}$$

$$-\frac{I_1}{Y_a} - \frac{I_1 + I_2}{Y_c} = 0 \quad \text{... (2)}$$

$$-I_1\left[\frac{1}{Y_a} + \frac{1}{Y_c}\right] = \frac{I_2}{Y_c}$$

$$-I_1\left[\frac{Y_a + Y_c}{Y_a Y_c}\right] = \frac{I_2}{Y_c}$$

$$I_1 = -\frac{I_2 \times Y_a}{Y_a + Y_c}$$

$$V_2 = -\frac{I_2 Y_a}{Y_c(Y_a + Y_c)} + \frac{I_2}{Y_c}$$

$$= \frac{I_2}{Y_c}\left[1 - \frac{Y_a}{Y_a + Y_c}\right]$$

$$= \frac{I_2}{Y_c}\left[\frac{Y_a + Y_c - Y_a}{Y_a + Y_c}\right]$$

$$\frac{I_2}{V_2} = Y_a + Y_c$$

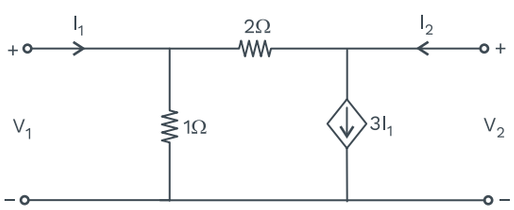
$$\left.\frac{I_2}{V_2}\right|_{V_1=0} = Y_a + Y_c = Y_{22}$$

$$\left.\frac{I_1}{V_2}\right|_{V_1=0} = -Y_a = Y_{12}$$

$$[Y]_\pi = \begin{bmatrix} Y_a + Y_b & -Y_a \\ -Y_a & Y_a + Y_c \end{bmatrix}$$

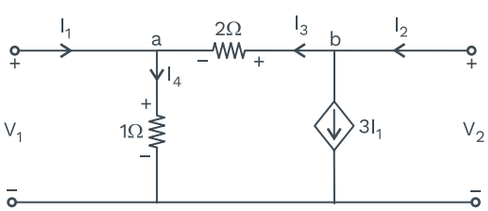
Example 1:

Find Z-Parameters.



Sol:

Method-1



Applying KCL at node 'b'

$I_2 = I_3 + 3I_1$

$I_3 = I_2 - 3I_1$ ...(1)

Applying KCL at node 'a'

$I_4 = I_3 + I_1 = I_2 - 3I_1 + I_1$

$I_4 = I_2 - 2I_1$ ...(2)

$V_1 = 1 \times I_4 = -2I_1 + I_2$

$V_1 = -2I_1 + I_2$

Here,

$Z_{11} = -2$, $Z_{12} = 1$

$I_4 \times 1 + I_3 \times 2 - V_2 = 0$

$V_2 = I_4 + 2I_3 = I_2 - 2I_1 + 2[I_2 - 3I_1] = I_2 - 2I_1 + 2I_2 - 6I_1$

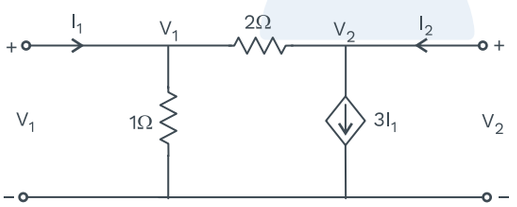
$V_2 = -8I_1 + 3I_2$

$Z_{21} = -8$, $Z_{22} = 3$

$$Z = \begin{bmatrix} -2 & 1 \\ -8 & 3 \end{bmatrix}$$

Method-2

Nodal Analysis:



Applying nodal at node 'V_1'

$$V_1\left[\frac{1}{1}+\frac{1}{2}\right] - V_2\left[\frac{1}{2}\right] = I_1$$

$$I_1 = V_1\left[\frac{3}{2}\right] + V_2\left[\frac{-1}{2}\right] \quad \text{... (1)}$$

$$Y_{11} = \frac{3}{2},\ Y_{12} = \frac{-1}{2}$$

Applying nodal at node 'V_2'

$$V_2\left[\frac{1}{2}\right] - \frac{V_1}{2} = I_2 - 3I_1 \quad \text{... (2)}$$

$$I_2 = 3I_1 - \frac{V_1}{2} + \frac{V_2}{2} = 3\left[\frac{3V_1}{2} - \frac{V_2}{2}\right] - \frac{V_1}{2} + \frac{V_2}{2} = V_1\left[\frac{9}{2} - \frac{1}{2}\right] + V_2\left[\frac{1}{2} - \frac{3}{2}\right]$$

$I_2 = V_1[4] + V_2[-1]$

$Y_{21} = 4, Y_{22} = -1$

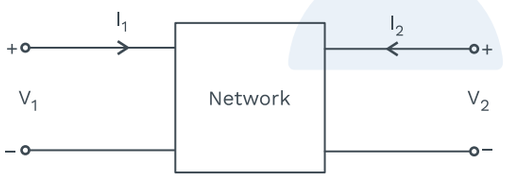
$$[Y] = \begin{bmatrix} \frac{3}{2} & \frac{-1}{2} \\ 4 & -1 \end{bmatrix}$$

$$[Z] = [Y]^{-1} = \frac{Adj[Y]}{|Y|} = \frac{\begin{bmatrix} -1 & \frac{1}{2} \\ -4 & \frac{3}{2} \end{bmatrix}}{-\frac{3}{2} + 2}$$

$$[Z] = \begin{bmatrix} -2 & 1 \\ -8 & 3 \end{bmatrix}$$

Example 2:

For the two-port network shown in the figure, find Z-Parameters.



$I_1 = 2V_1 + V_2$

$I_2 = V_1 + V_2$

Sol:

I^st Method:

$$Y = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

$$[Z] = [Y]^{-1} = \frac{Adj[Y]}{|Y|} = \frac{\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}}{2-1}$$

$$[Z] = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$$

Grey Matter Alert!

$$Y_{in} = \frac{I_1}{V_1} = \frac{1}{Z_{in}}$$

$$Y_{out} = \frac{I_2}{V_2} = \frac{1}{Z_{out}}$$

IInd Method:

$V_1 = Z_{11} I_1 + Z_{12} I_2$

$V_2 = Z_{21} I_1 + Z_{22} I_2$

Case 1

$I_2 = 0$

$V_1 + V_2 = 0$

$V_1 = - V_2$

$I_1 = 2V_1 - V_1$

$I_1 = V_1$

$$\left.\frac{V_1}{I_1}\right|_{I_2=0} = 1 = Z_{11}$$

$I_1 = 2V_1 + V_2 = -2V_2 + V_2$

$I_1 = - V_2$

$$\left.\frac{V_2}{I_1}\right|_{I_2=0} = -1 = Z_{21}$$

Case 2

$I_1 = 0$

$2V_1 + V_2 = 0$

$V_2 = -2V_1$

$I_2 = V_1 + V_2$

$$I_2 = V_2 - \frac{V_2}{2}$$

$$\left.\frac{V_2}{I_2}\right|_{I_1=0} = 2 = Z_{22}$$

$I_2 = V_1 + V_2$

$I_2 = V_1 - 2V_1$

$I_2 = -V_1$

$$\left.\frac{V_1}{I_2}\right|_{I_1=0} = -1 = Z_{12}$$

$Z_{11} = 1$

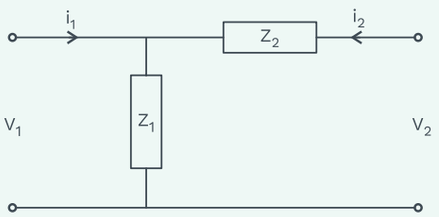
$Z_{12} = -1$

$Z_{21} = -1$

$Z_{22} = 2$

Previous Years' Question

In the two port networks shown in the figure, the Z-matrix is given by:



a) $\begin{bmatrix} Z_1 & Z_1 + Z_2 \\ Z_1 + Z_2 & Z_2 \end{bmatrix}$

b) $\begin{bmatrix} Z_1 & Z_1 \\ Z_1 + Z_2 & Z_2 \end{bmatrix}$

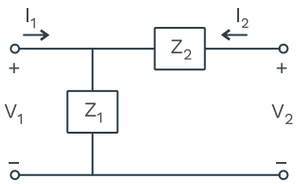
c) $\begin{bmatrix} Z_1 & Z_2 \\ Z_2 & Z_1 + Z_2 \end{bmatrix}$

d) $\begin{bmatrix} Z_1 & Z_1 \\ Z_1 & Z_1 + Z_2 \end{bmatrix}$

$$[Z] = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$$

Example 3:

For the two-port network shown in the figure, the Z-matrix is _____.



Sol:

Applying KVL in input and output loops, we get:

$V_1 - (I_1 + I_2)\, Z_1 = 0$

$V_1 = Z_1 I_1 + Z_1 I_2$ ...(1)

Similarly,

$V_2 - I_2 Z_2 - (I_1 + I_2)\, Z_1 = 0$

$V_2 = Z_1 I_1 + (Z_1 + Z_2)\, I_2$ ...(2)

We know that, standard equations of Z-parameters

$V_1 = Z_{11} I_1 + Z_{12} I_2$ ...(3)

$V_2 = Z_{21} I_1 + Z_{22} I_2$ ...(4)

Comparing equation 1) and equation 3), we get:

$Z_{11} = Z_1$

$Z_{12} = Z_1$

Comparing equation 2) and equation 4), we get:

$Z_{21} = Z_1$

$Z_{22} = Z_1 + Z_2$

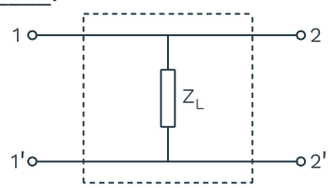
So, Z-matrix is:

$$[Z] = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix}$$

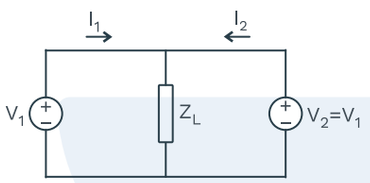
$$[Z] = \begin{vmatrix} Z_1 & Z_1 \\ Z_1 & Z_1 + Z_2 \end{vmatrix}$$

Example 4:

The Z-parameter matrix of the single element two port network of the figure shown below is _____.



Sol:



From the circuit,

$$\frac{V_1}{Z_L} = I_1 + I_2 \quad \text{... (1)}$$

$$V_1 = Z_L I_1 + Z_L I_2 \quad \text{... (2)}$$

We know that,

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \quad \text{... (3)}$$

From equations 2) and 3),

$$Z_{11} = Z_L$$

$$Z_{12} = Z_L$$

Again, substituting $V_1 = V_2$ in equation 1),

$$\frac{V_2}{Z_L} = I_1 + I_2 \quad \text{... (4)}$$

$$V_2 = Z_L I_1 + Z_L I_2 \quad \text{... (5)}$$

We know that,

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \quad \text{... (6)}$$

From equations 5) and 6)

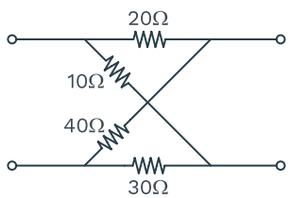
$$Z_{21} = Z_L$$

$$Z_{22} = Z_L$$

$$Z = \begin{bmatrix} Z_L & Z_L \\ Z_L & Z_L \end{bmatrix} \Omega$$

Example 5:

For the given two-port resistive network, the Z-parameter Z_{11} is _____.

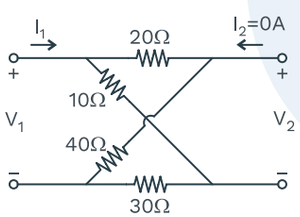


Grey Matter Alert!

For the existence of h-parameters, $I_1 \neq f(V_2)$, $V_2 \neq f(I_1)$

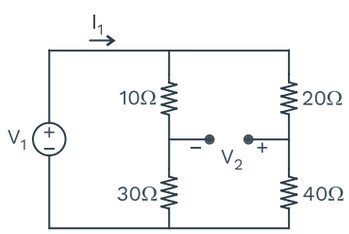
Sol:

Given,



$$Z_{11} = \left.\frac{V_1}{I_1}\right|_{I_2=0}$$ (Open circuit output terminal)

A simplified circuit is shown below.



$V_1 = I_1[(10 + 30)] \parallel [(20 + 40)]$

$V_1 = I_1[40 \parallel 60]$

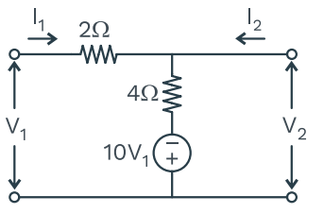
$$Z_{11} = \frac{V_1}{I_1} = 24\Omega$$

Rack Your Brain

Which two-port network is used in transistor modelling and why?

Example 6:

Find the Z-parameters, Z_{11} and Z_{21}, for the two-port network shown in the figure.



Sol:

Applying KVL in input loop,

$V_1 = 2I_1 + 4(I_1 + I_2) - 10V_1$

or $V_1 = \frac{6}{11}I_1 + \frac{4}{11}I_2$ ... (1)

We know that,

$V_1 = Z_{11}I_1 + Z_{12}I_2$ ... (2)

Comparing equations 1) and 2), we get:

Thus, $Z_{11} = \frac{6}{11}\Omega$

Applying KVL in output loop:

$V_2 = 4(I_1 + I_2) - 10V_1$

$= 4(I_1 + I_2) - 10\left(\frac{6I_1}{11} + \frac{4I_2}{11}\right)$

$V_2 = -\frac{16}{11}I_1 + \frac{4}{11}I_2$ ... (3)

We know that,

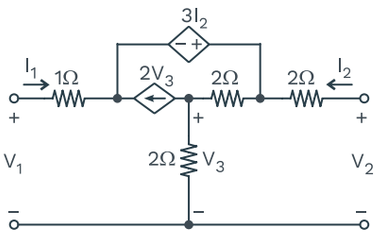
$V_2 = Z_{21}I_1 + Z_{22}I_2$ ... (4)

Comparing equations 3) and 4), we get:

Thus, $Z_{21} = -\frac{16}{11}\Omega$

Example 7:

For the circuit shown below, the input impedance $Z_{11} = \left.\frac{V_1}{I_1}\right|_{I_2=0}$ is _____.

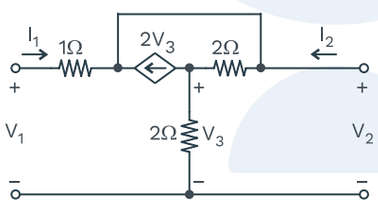


Sol:

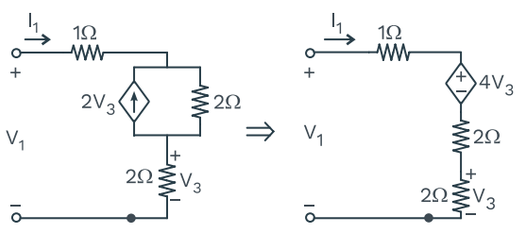
Given,

$$Z_{11} = \left.\frac{V_1}{I_1}\right|_{I_2=0}$$

When $I_2 = 0$ A, dependent voltage source $3I_2 = 0$ and it acts as short circuit.



We can see that the dependent current source and the rightmost 2Ω resistor (the horizontal one) are in parallel, so a simplified circuit is shown below.



Applying KVL in the loop,

$V_1 - 1I_1 - 4V_3 - 2I_1 - 2I_1 = 0$

$V_1 - 1I_1 - 4(2I_1) - 2I_1 - 2I_1 = 0 \qquad (V_3 = 2I_1)$

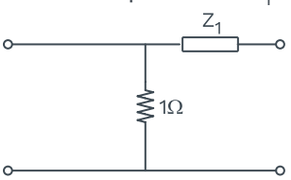
$V_1 - 13I_1 = 0$

$$Z_{11} = \frac{V_1}{I_1} = 13\Omega$$

Example 8:

For the network shown in the figure, the transmission parameter matrix is $\begin{bmatrix} 1 & -j1\Omega \\ 1S & (1-j1) \end{bmatrix}$.

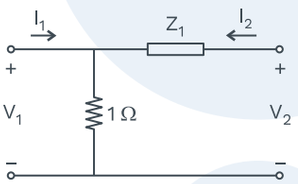
What is the value of impedance Z_1?



Sol:

Given,

Transmission parameters are obtained as:



$$B = -\left.\frac{V_1}{I_2}\right|_{V_2=0}$$

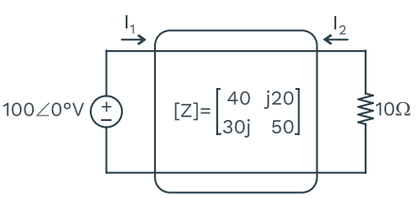
If $V_2 = 0$, $I_2 = \dfrac{0 - V_1}{Z_1}$

$$B = -\frac{V_1}{I_2} = Z_1$$

So, $Z_1 = -j1$

Example 9:

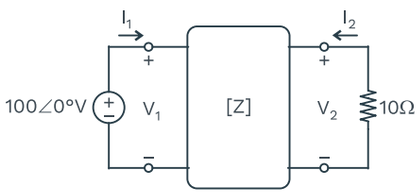
In the given two ports network, what are the values of current I_1 and I_2?



Rack Your Brain

Which two-port parameter is used in the power system and why?

Sol:



Z-parameter equation for the network is:

$V_1 = Z_{11}I_1 + Z_{12}I_2$ ... (1)

$V_2 = Z_{21}I_1 + Z_{22}I_2$ ... (2)

$V_1 = 100\angle 0°V$, $V_2 = -10I_2$

Substituting V_1 and V_2 into equation 1) and 2), we get:

So, $100 = 40I_1 + j20I_2$ ... (3)

$-10I_2 = j30I_1 + 50I_2$

$-60I_2 = j30I_1 \Rightarrow I_2 = I_1/j2$

Substituting I_2 into equation 3) and we get:

$$100 = 40I_1 + j20\frac{I_1}{j2}$$

$100 = (40 + 10)\ I_1$

$I_1 = 2\angle 0°$ A

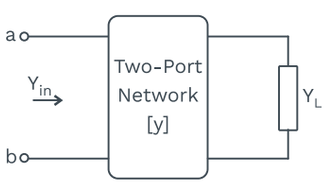
$$I_2 = \frac{2\angle 0°}{j2} = 1\angle -90° \text{ A}$$

Example 10:

The short circuit admittance parameter of the given two port network is:

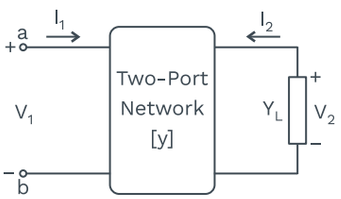
$$[y] = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix}$$

The input admittance seen from terminal a-b is_____.



Sol:

Given,



$$Y_{in} = \frac{I_1}{V_1}$$

For the two-port network,

$$I_1 = y_{11}V_1 + y_{12}V_2 \quad \text{... (1)}$$

$$I_2 = y_{21}V_1 + y_{22}V_2 \quad \text{... (2)}$$

Also, $I_2 = -V_2Y_L$

Substituting I_2 into equation 2),

$$-V_2Y_L = y_{21}V_1 + y_{22}V_2$$

$$V_2(y_{22} + Y_L) = -y_{21}V_1$$

$$V_2 = -\frac{y_{21}}{y_{22} + Y_L}V_1$$

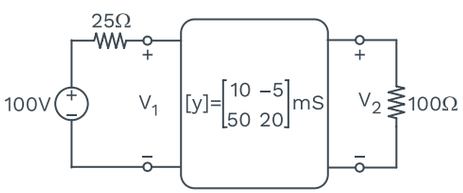
Substituting V_2 into equation 1),

$$I_1 = y_{11}V_1 + y_{12}\left(-\frac{y_{21}}{y_{22} + Y_L}\right)V_1 = V_1\left[y_{11} - \frac{y_{12}y_{21}}{y_{22} + Y_L}\right]$$

$$Y_{in} = \frac{I_1}{V_1} = y_{11} - \left(\frac{y_{12}y_{21}}{y_{22} + Y_L}\right)$$

Example 11:

For the circuit shown below, the voltage V_1 is _____.



Sol:

Given,

$I_1 = y_{11}V_1 + y_{12}V_2$

$I_1 = 10 \times 10^{-3}\, V_1 - 5 \times 10^{-3}V_2$ ... (1)

Since $100 = 25I_1 + V_1$ (KVL in the input loop)

Substituting I_1 from equation 1) into above equation,

$100 = 25(10 \times 10^{-3}\, V_1 - 5 \times 10^{-3}V_2) + V_1 = 0.25V_1 - 0.125V_2 + V_1$

$800 = 10V_1 - V_2$ ... (2)

$I_2 = y_{21}V_1 + y_{22}V_2$

$I_2 = 50 \times 10^{-3}\, V_1 + 20 \times 10^{-3}V_2$ ... (3)

Since $V_2 = -100I_2$ (Ohm's Law)

Substituting I_2 from equation 3) into above equation,

$V_2 = -5V_1 - 2V_2$

$3V_2 + 5V_1 = 0$ ... (4)

Solving equation 2) and 4), we get-

$V_1 = 68.6V$

Hybrid (h) parameters:

$V_1 = f(I_1, V_2)$

$I_2 = f(I_1, V_2)$

$V_1 = h_{11}\, I_1 + h_{12}\, V_2$

$I_2 = h_{21}\, I_1 + h_{22}\, V_2$

V_1 and I_2 are dependent variables

I_1 and V_2 are independent variables

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$

$$[h] = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix}$$

$$[h]^{-1} = \frac{\text{Adj}[h]}{\Delta h} = \frac{\begin{bmatrix} h_{22} & -h_{12} \\ -h_{21} & h_{11} \end{bmatrix}}{[h_{11}h_{22} - h_{12}h_{21}]} = [g]$$

$$h_{11} \neq \frac{1}{g_{11}};\ h_{12} \neq \frac{1}{g_{12}}$$

$$h_{21} \neq \frac{1}{g_{21}};\ h_{12} \neq \frac{1}{g_{22}}$$

Because conditions for both parameters are different.

Calculation of h-parameters:

Case 1 [V_2 = 0V]

$$h_{11} = \left.\frac{V_1}{I_1}\right|_{V_2=0}$$ = Input impedance, when V_2 = 0 V.

$$h_{21} = \left.\frac{I_2}{I_1}\right|_{V_2=0}$$ = Forward current gain, when V_2 = 0 V.

Case 2 [I_1 = 0A]

$$h_{22} = \left.\frac{I_2}{V_2}\right|_{I_1=0}$$ = Output admittance, when I_1 = 0A

$$h_{12} = \left.\frac{V_1}{V_2}\right|_{I_1=0}$$ = Reverse voltage gain, when I_1 = 0A

h-parameters using [Z] and [Y] parameters:

Using Z-parameters	Using Y-parameters
$V_1 = Z_{11} I_1 + Z_{12} I_2$	$I_1 = Y_{11} V_1 + Y_{12} V_2$
$V_2 = Z_{21} I_1 + Z_{22} I_2$	$I_2 = Y_{21} V_1 + Y_{22} V_2$
$h_{11} = \frac{\Delta Z}{Z_{22}}$	$h_{11} = \frac{1}{Y_{11}}$
$h_{12} = \frac{Z_{12}}{Z_{22}}$	$h_{12} = -\frac{Y_{12}}{Y_{11}}$
$h_{21} = -\frac{Z_{21}}{Z_{22}}$	$h_{22} = \frac{Y_{21}}{Y_{11}}$
$h_{22} = \frac{1}{Z_{22}}$	$h_{22} = \frac{\Delta Y}{Y_{11}}$
$\Delta Z = Z_{11} Z_{22} - Z_{12} Z_{21}$	$\Delta Y = Y_{11} Y_{22} - Y_{12} Y_{21}$

Table 1.1 Comparison Between Z- Parameters and Y- Parameters

Modelling of h-parameters:

$V_1 = h_{11} I_1 + h_{12} V_2 \rightarrow$ KVL

$I_2 = h_{21} I_1 + h_{22} V_2 \rightarrow$ KCL

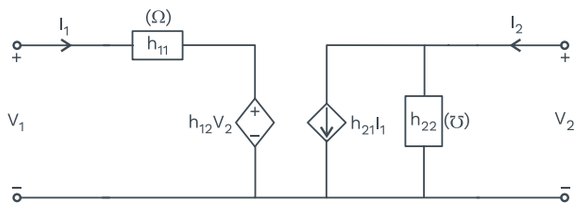


Fig. 4.18 Representation of h- Parameters

Symmetrical $\rightarrow \Delta h = h_{11}h_{22} - h_{12}h_{21} = 1$

Reciprocal $\rightarrow h_{12} = -h_{21}$

g-parameters:

$I_1 = f(V_1, I_2)$

$V_2 = f(V_1, I_2)$

$I_1 = g_{11} V_1 + g_{12} I_2$

$V_2 = g_{21} V_1 + g_{22} I_2$

I_1 and $V_2 \rightarrow$ Dependent variables

V_1 and $I_2 \rightarrow$ Independent Variable.

$$\begin{bmatrix} I_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ I_2 \end{bmatrix}; \quad [g] = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix}$$

$$[g]^{-1} = \frac{\text{Adj}[g]}{\Delta g} = \frac{\begin{bmatrix} g_{22} & -g_{12} \\ -g_{21} & g_{11} \end{bmatrix}}{[g_{11} \times g_{22} - g_{12} \times g_{21}]} = [h]$$

$$\begin{bmatrix} g_{11} \neq \frac{1}{h_{11}} & g_{12} \neq \frac{1}{h_{12}} \\ g_{21} \neq \frac{1}{h_{21}} & g_{22} \neq \frac{1}{h_{22}} \end{bmatrix}$$

Grey Matter Alert!

For the existence of g-parameters, $V_1 \neq f(I_2)$ and $I_2 \neq f(V_1)$

Grey Matter Alert!

As V_1 and I_1 are the variables at the same port, hence the electrical modelling of ABCD is not possible.

Modelling of g-parameters:

$I_1 = g_{11} V_1 + g_{12} I_2$

$V_2 = g_{21} V_1 + g_{22} I_2$

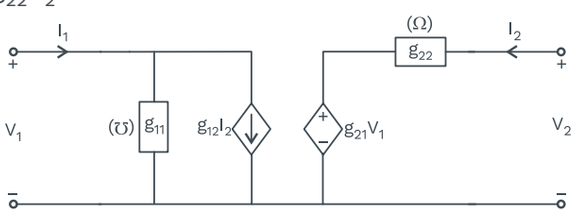


Fig. 4.19 Representation of g- Parameters

Symmetrical $\rightarrow \Delta g = g_{11}g_{22} - g_{12}g_{21} = 1$
Reciprocal $\rightarrow g_{12} = -g_{21}$

A B C D (T)-parameters:

- They are used for the analysis of transmission lines.
- Hence, they are also known as transmission line parameters [T-parameters]
- $V_1 = f(V_2, I_2)$
  $I_1 = f(V_2, I_2)$
  V_1 and I_1 are dependent variables.

V_2 and I_2 are independent variables.

$V_1 = AV_2 - BI_2$

$I_1 = CV_2 - DI_2$

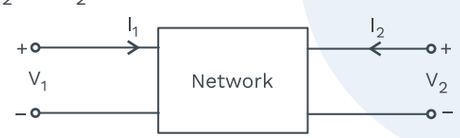


Fig. 4.20

$V_1 = AV_2 + BI_2$

$I_1 = CV_2 + DI_2$



Fig. 4.21

Case 1 [$V_2 = 0V$]

$$B = -\frac{V_1}{I_2} \Rightarrow (\Omega)$$

$$D = -\frac{I_1}{I_2} \Rightarrow \text{Unitless}$$

Case 2 [$I_2 = 0A$]

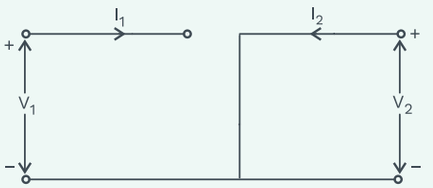
$$A = \frac{V_1}{V_2} \Rightarrow \text{Unitless}$$

$$C = \frac{I_1}{V_2} \Rightarrow \Omega^{-1} \text{ or } S$$

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

Previous Years' Question

The parameter type and the matrix representation of the relevant two port parameters that describe the circuit shown are:



a) z-parameters, $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

b) h-parameters, $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

c) g-parameters, $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

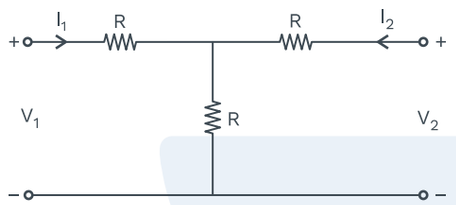
d) z-parameters, $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$$[ABCD] = [T] = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

Reciprocal $\rightarrow \Delta ABCD = 1 = AD - BC$
Symmetrical $\rightarrow [A = D]$

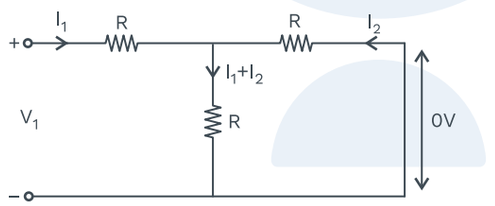
Example 12:

Find h_{21} for the given two-port network.



Sol:

For finding h_{21} $[V_2 = 0V]$



$$\left.\frac{V_1}{I_1}\right|_{V_2=0} = R + \frac{R}{2} = \frac{3R}{2} = h_{11}$$

$(I_1 + I_2)\, R + I_2\, R = 0$

$I_1R + I_2\,(2R) = 0$

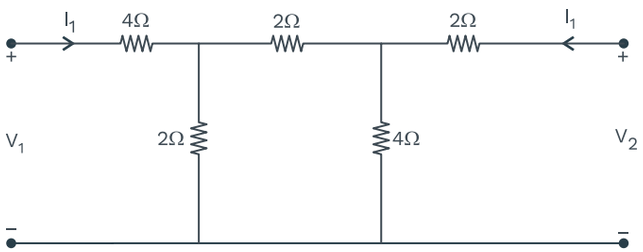
$I_2(2R) = -I_1R$

$$\frac{I_2}{I_1} = \frac{-R}{2R} = -\frac{1}{2}$$

$$\left.\frac{I_2}{I_1}\right|_{V_2=0} = -\frac{1}{2} = h_{21}$$

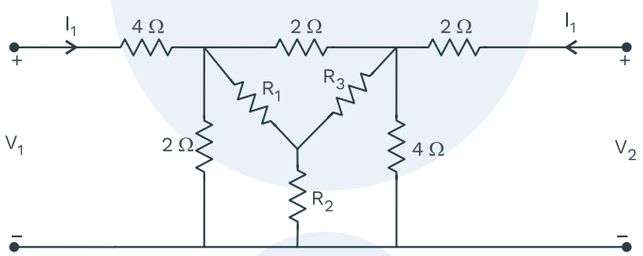
Example 13:

Find h- parameters for the given two-port network.



Sol:

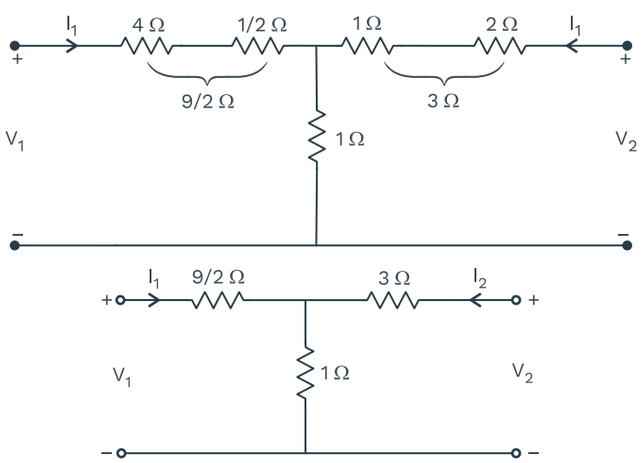
Converting delta into the star as shown in below figure.



$$R_1 = \frac{2 \times 2}{2+2+4} = \frac{4}{8} = \frac{1}{2}\Omega$$

$$R_2 = \frac{2 \times 4}{2+2+4} = 1\Omega$$

$$R_3 = \frac{4 \times 2}{2+2+4} = 1\Omega$$

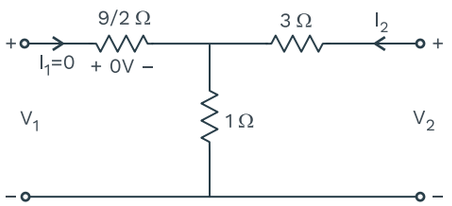


Standard equations of h-parameters are:

$V_1 = h_{11} I_1 + h_{12} V_2$

$I_2 = h_{21} I_1 + h_{22} V_2$

Case 1 [I_1 = 0V]



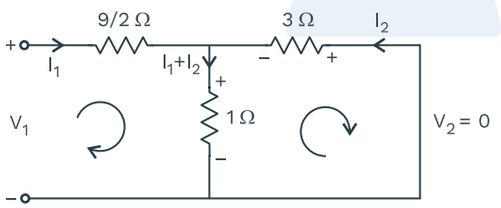
$V_1 = I_2 \times 1$ ... (1)

$V_2 = I_2 \times 4$ ... (2)

$$\left.\frac{I_2}{V_2}\right|_{I_1=0} = \frac{1}{4} = h_{22}$$

$$\left.\frac{V_1}{V_2}\right|_{I_1=0} = \frac{I_2 \times 1}{I_2 \times 4} = \frac{1}{4} = h_{12}$$

Case 2 [V_2 = 0V]



$(I_1 + I_2) \times 1 + I_2 \times 3 = 0$

$I_1 + I_2 \times 4 = 0$

$$V_1 = I_1 \times \frac{9}{2} + (I_1 + I_2) \times 1$$

$$\left.\frac{I_2}{I_1}\right|_{V_2=0} = \frac{-1}{4} = h_{21}$$

$$V_1 = I_1 \times \left(\frac{9}{2} + 1\right) + I_2$$

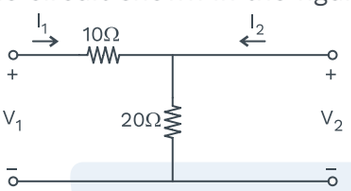
$$V_1 = \left(\frac{9}{2} + 1\right) I_1 + \left(-\frac{I_1}{4}\right)$$

$$V_1 = I_1\left(\frac{9}{2} + 1 - \frac{1}{4}\right)$$

$$\left.\frac{V_1}{I_1}\right|_{V_2=0} = \frac{21}{4} = h_{11}$$

Example 14:

The h-parameters for the circuit shown in the figure are _____.



Sol:

Given,

h-parameters are described by the following equations:

$V_1 = h_{11}I_1 + h_{12}V_2$ ... (1)

$I_2 = h_{21}I_1 + h_{22}V_2$ ... (2)

Applying KVL in the input loop:

$V_1 = 10I_1 + V_2$ ... (3)

Applying KVL in the output loop,

$$\frac{V_2}{20} = I_1 + I_2$$

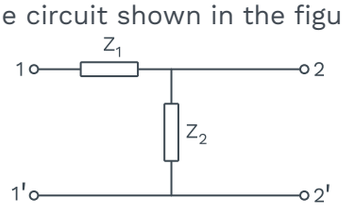
$$\text{or } I_2 = -I_1 + \frac{V_2}{20} \quad \text{... (4)}$$

Comparing equations 3) and 4) with equations 1) and 2), We get:

$$\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} 10\Omega & 1 \\ -1 & 0.05\,S \end{bmatrix}$$

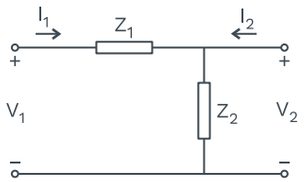
Example 15:

The h-parameters for the circuit shown in the figure are _____.



Sol:

Given,



h-parameters are defined by following equations:

$V_1 = h_{11}I_1 + h_{12}V_2$ ... (1)

$I_2 = h_{21}I_1 + h_{22}V_2$ ... (2)

$$I_1 = \frac{V_1 - V_2}{Z_1}$$

$V_1 = Z_1I_1 + V_2$ ... (3)

$$I_1 + I_2 = \frac{V_2}{Z_2}$$

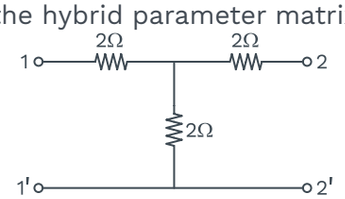
$$I_2 = -I_1 + \frac{1}{Z_2}V_2$$ ... (4)

Comparing equations 3) and 4) with equations 1) and 2) respectively, we get:

$$[h] = \begin{bmatrix} Z_1 & 1 \\ -1 & 1/Z_2 \end{bmatrix}$$

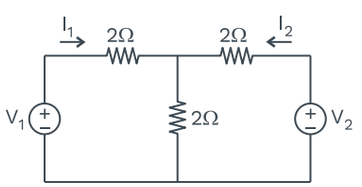
Example 16:

For the given network, the hybrid parameter matrix is:

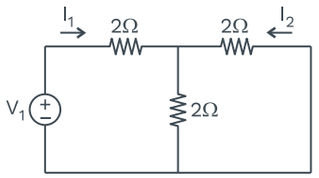


Sol:

Given,



$$h_{11} = \left.\frac{V_1}{I_1}\right|_{V_2=0}$$ (Short circuit output terminal)



$V_1 = I_1[2 + (2 \,||2)]$

$V_1 = I_1[2 + 1]$

$$h_{11} = \frac{V_1}{I_1} = 3\Omega$$

$$h_{21} = \left.\frac{I_2}{I_1}\right|_{V_2=0}$$ (Same circuit as shown above)

$$I_2 = \frac{2}{2+2}(-I_1)$$ (Using current division)

$$h_{21} = \frac{I_2}{I_1} = -\frac{1}{2}$$

$$h_{12} = \left.\frac{V_1}{V_2}\right|_{I_1=0}$$ (Open circuit output terminal)

$$V_1 = \left(\frac{2}{2+2}\right)V_2$$

$$h_{12} = \frac{V_1}{V_2}$$

$$h_{12} = \frac{1}{2}$$

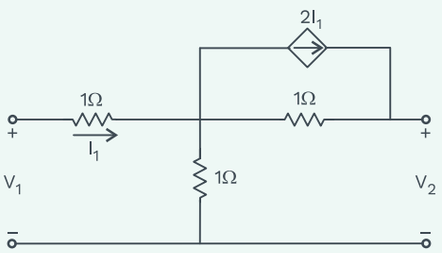
$$h_{22} = \left.\frac{I_2}{V_2}\right|_{I_1=0}$$ (Same circuit as shown above)

$V_2 = I_2(2 + 2)$

$$h_{22} = \frac{I_2}{V_2} = \frac{1}{4}S$$

Previous Years' Question

The two-port network shown the h_{11} parameter where $h_{11} = \frac{V_1}{I_1}$, when $V_2 = 0$ in ohms is ..... (up to 2 decimal places).



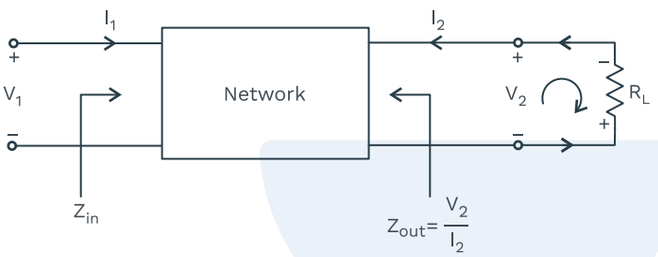
Example 17:

A two-port network is represented by:

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

If port-2 is terminated by R_L, then the input impedance seen at port-1 will be _____.

Sol:



The standard equation of T-parameters are:

$$V_1 = AV_2 - BI_2 \quad \text{..... (1)}$$

$$I_1 = CV_2 - DI_2 \quad \text{..... (2)}$$

$$Z_{in} = \frac{V_1}{I_1}$$

$$V_2 = -I_2 R_L$$

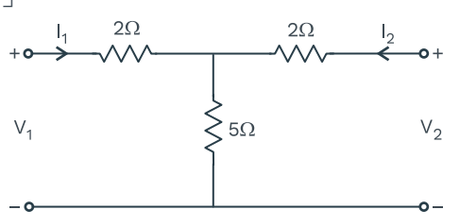
$$\frac{V_1}{I_1} = \frac{AV_2 - BI_2}{CV_2 - DI_2} = \frac{-AI_2R_L - BI_2}{-CI_2R_L - DI_2}$$

$$\frac{V_1}{I_1} = \frac{AR_L + B}{CR_L + D} = Z_{in}$$

Example 18:

For the two-port network given below, find the ABCD parameters.

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$



Sol:

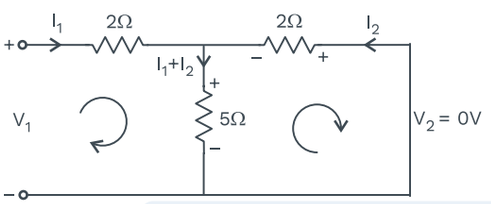
We know that,

Standard equations of ABCD-parameters are:

$V_1 = AV_2 - BI_2$

$I_1 = CV_2 - DI_2$

Case-1 [V_2 = 0V]



$5(I_1 + I_2) + 2I_2 = 0$

$5I_1 + 7I_2 = 0$

$$\left.\frac{I_1}{I_2}\right|_{V_2=0} = \left(-\frac{7}{5}\right)$$

$$D = \frac{7}{5}$$

$V_1 = 7I_1 + 5I_2$

$$V_1 = 7\left(-\frac{7}{5}\right)I_2 + 5I_2$$

$$\frac{V_1}{I_2} = -\frac{7\times 7}{5} + 5 = -\frac{49}{5} + 5$$

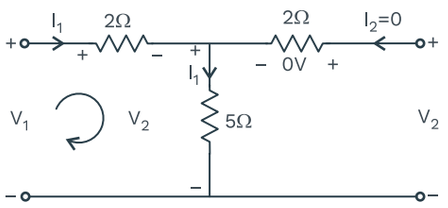
$$\frac{V_1}{I_2} = -\frac{24}{5}$$

$$B = -\frac{V_1}{I_2} = -\left(-\frac{24}{5}\right)$$

$$B = \frac{24}{5}$$

Case-2 [I_2 = 0A]

$V_1 - I_1 \times 2 - V_2 = 0$ ...(1)



$$V_2 = I_1 \times 5 \qquad \ldots(2)$$

$$\left.\frac{I_1}{V_2}\right|_{I_2=0} = \frac{1}{5} = C$$

From equation 1),

$$V_1 - V_2 = I_1 \times 2$$

$$V_1 = V_2 + I_1 \times 2$$

$$V_1 = V_2 + \frac{V_2}{5} \times 2$$

$$V_1 = V_2\left[1 + \frac{2}{5}\right]$$

$$V_1 = V_2\left[\frac{7}{5}\right]$$

$$\left.\frac{V_1}{V_2}\right|_{I_2=0} = \frac{7}{5} = A$$

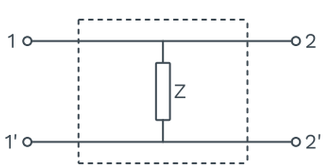
$$[T] = \begin{bmatrix} \frac{7}{5} & \frac{24}{5} \\ \frac{1}{5} & \frac{7}{5} \end{bmatrix}$$

Rack Your Brain

Which ABCD parameter is dimensionless?

Example 19:

The transmission parameters matrix of the single element two-port circuit of the figure below is _____.



Sol:

Transmission parameters are defined as:

$V_1 = AV_2 - BI_2$ ...(1)

$I_1 = CV_2 - DI_2$ ...(2)

In the circuit,

$V_1 = V_2$

$V_1 = 1V_2 + 0I_2$ ...(3)

$$I_1 + I_2 = \frac{V_2}{Z}$$

$$I_1 = \frac{1}{Z}V_2 - I_2$$ ...(4)

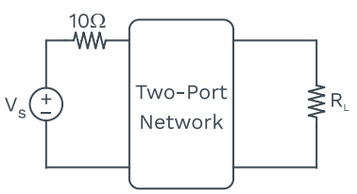
Comparing equations 3) and 4) with equations 1) and 2) respectively, we get:

A = 1, B = 0, C = 1/Z, D = 1

$$[T] = \begin{bmatrix} 1 & 0 \\ 1/Z & 1 \end{bmatrix}$$

Example 20:

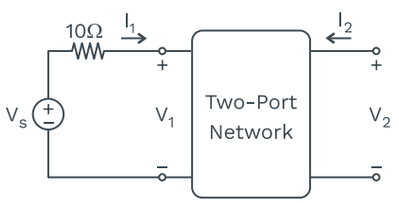
In the circuit shown below ABCD parameter matrix of the two-port network is $\begin{bmatrix} 4 & 20\Omega \\ 0.1S & 2 \end{bmatrix}$.



What should be the value of load R_L to transfer the maximum power?

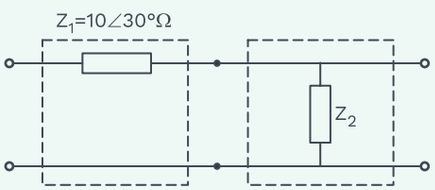
Sol:

Given,



Previous Years' Question

Two networks are connected in cascade as shown in the figure. With the usual notations, the equivalent A, B, C and D constants are obtained. Given that, C= 0.25∠45°, the value of Z_2 is:



a) 10∠30° Ω
b) 40∠−45°Ω
c) 1Ω
d) 0Ω

ABCD parameters are given as:

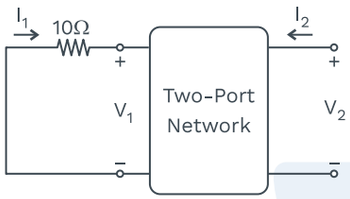
$V_1 = AV_2 - BI_2$

$I_1 = CV_2 - DI_2$

$V_1 = 4V_2 - 20I_2$ ... (1)

$I_1 = 0.1V_2 - 2I_2$ ... (2)

To obtain maximum power transfer, we must calculate Thevenin's impedance across load terminal.



So, $Z_{Th} = \frac{V_2}{I_2}$

Substituting $V_1 = -10\ I_1$ in equation 1),

$-10\ I_1 = 4V_2 - 20I_2$ ... (3)

Substituting I_1 from equation (2) in equation (3), we get-

$-10(0.1V_2 - 2I_2) = 4V_2 - 20I_2$

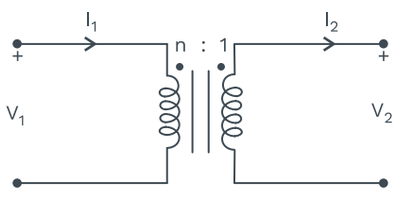
$-V_2 + 20I_2 = 4V_2 - 20I_2$

$40\ I_2 = 5V_2$

$$Z_{Th} = \frac{V_2}{I_2} = \frac{40}{5} = 8\,\Omega$$

So, $R_L = 8\Omega$ (For maximum power transfer)

Transformer:

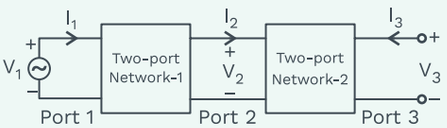


$$\frac{V_1}{V_2} = \frac{N_1}{N_2} = n = a$$

$$\frac{I_2}{I_1} = \frac{N_1}{N_2} = n = a$$

Previous Years' Question

Two passive two port networks are connected in cascade, as shown in the figure. A voltage source is connected at port 1.



Given,

A_1, B_1, C_1, D_1, A_2, B_2, C_2 and D_2 are the generalized circuit at port 3 consisting of a voltage source V_T and an impedance Z_T, connected in series, then:

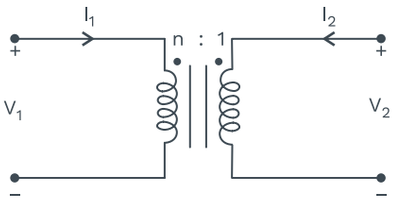
a) $V_T = \frac{V_1}{A_1A_2}, Z_T = \frac{A_1B_2 + B_1D_2}{A_1A_2 + B_1C_2}$

b) $V_T = \frac{V_1}{A_1A_2 + B_1C_2}, Z_T = \frac{A_1B_2 + B_1D_2}{A_1A_2}$

c) $V_T = \frac{V_1}{A_1A_2}, Z_T = \frac{A_1B_2 + B_1D_2}{A_1 + A_2}$

d) $V_T = \frac{V_1}{A_1A_2 + B_1C_2}, Z_T = \frac{A_1B_2 + B_1D_2}{A_1A_2 + B_1C_2}$

- The transformer is represented as a two-port network in the network theory, shown below:



$$\frac{V_1}{V_2} = n = \frac{N_1}{N_2} \rightarrow \{V_1 = nV_2\}$$

$$-\frac{I_2}{I_1} = n = \frac{N_1}{N_2} \rightarrow \{I_2 = -nI_1\}$$

or $\left\{I_1 = -\frac{I_2}{n}\right\}$

As, $V_1 = f(V_2) \rightarrow$ Y-Parameter will never exist

$I_2 = f(I_1) \rightarrow$ Z-Parameter will never exist

h-parameter:

$V_1 = h_{11} I_1 + h_{12} V_2$

$I_2 = h_{21} I_1 + h_{22} V_2$

$V_1 = 0I_1 + nV_2$

$I_2 = -nI_1 + 0V_2$

$$h = \begin{bmatrix} 0 & n \\ -n & 0 \end{bmatrix}$$

$h_{12} = -h_{21} \rightarrow$ Reciprocal

$\Delta h = n^2 \neq 1 \rightarrow$ Asymmetrical

- **[ABCD] – parameters:**

We know that,

$V_1 = AV_2 - BI_2$

$I_1 = CV_2 - DI_2$

$V_1 = nV_2 - 0 \times I_2$

$$I_1 = 0 \times V_2 - \frac{1}{n} I_2$$

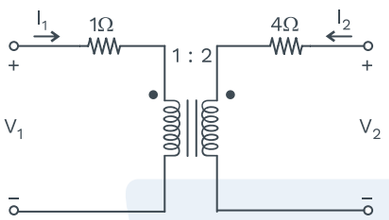
$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} n & 0 \\ 0 & \frac{1}{n} \end{bmatrix}$$

$\Delta T = n \times \frac{1}{n} - 0 = 1 \rightarrow$ Reciprocal

$A \neq D \rightarrow$ Not symmetrical

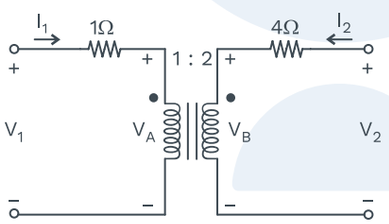
Example 21:

For the given two-port network, the h-parameters are _____.



Sol:

Given,



For an ideal transformer,

$$\frac{V_A}{V_B} = \frac{1}{n} = \frac{1}{2}$$

$$\frac{I_1}{I_2} = -n = -2$$

Applying KVL in primary and secondary loops,

$V_1 - 1I_1 - V_A = 0$ ...(1)

$V_2 - 4I_2 - V_B = 0$ ...(2)

Substituting $V_A = V_B/2$ into equation 1):

$$V_1 - I_1 - \frac{V_B}{2} = 0 \quad \text{...(3)}$$

From equation 2), put $V_B = (V_2 - 4I_2)$ in equation 3), we get:

$$V_1 - I_1 - \frac{(V_2 - 4I_2)}{2} = 0$$

$$2V_1 - 2I_1 - V_2 + 4I_2 = 0$$

$$2V_1 - V_2 - 2I_1 + 4\left(-\frac{I_1}{2}\right) = 0 \quad \left\{I_2 = \left(-\frac{I_1}{2}\right)\right\}$$

$$2V_1 = V_2 + 4I_1$$

$$V_1 = \frac{1}{2}V_2 + 2I_1$$

$$V_1 = 2I_1 + 0.5V_2$$

$h_{11} = 2\Omega$, $h_{12} = 0.5$

Since, $I_2 = -\frac{I_1}{2}$

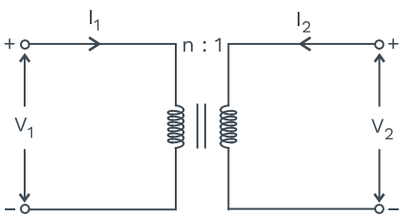
or, $I_2 = -0.5I_1 + 0V_2$

So, $h_{21} = -0.5$, $h_{22} = 0$

$$[h] = \begin{bmatrix} 2\Omega & 0.5 \\ -0.5 & 0\,S \end{bmatrix}$$

Example 22:

The ABCD parameters of an ideal n : 1 transformer shown in the figure are $\begin{bmatrix} n & 0 \\ 0 & x \end{bmatrix}$.



The value of x will be_____.

Sol:

For given transformer,

$V_1 = nV_2 + (0)(-I_2)$

$I_1 = 0\,V_2 + \left(\frac{+1}{n}\right)(-I_2)$

Comparing with standard equations:

$V_1 = AV_2 - BI_2$

$I_1 = CV_2 - DI_2$

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} n & 0 \\ 0 & \frac{1}{n} \end{bmatrix}$$

$$x = \frac{1}{n}$$

INTERCONNECTION OF TWO PORT NETWORKS:

i) Series-interconnection:

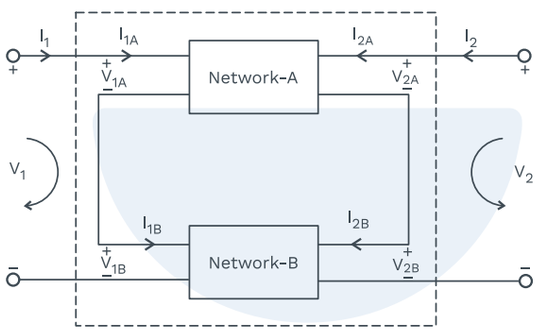


Fig. 4.22

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

$[Z] = [Z]_A + [Z]_B$

For network-A

$V_{1A} = Z_{11A}\, I_{1A} + Z_{12A}\, I_{2A}$

$V_{2A} = Z_{21A}\, I_{1A} + Z_{22A}\, I_{2A}$

For network-B

$V_{1B} = Z_{11B}\, I_{1B} + Z_{12B}\, I_{2B}$

$V_{2B} = Z_{21B}\, I_{1B} + Z_{22B}\, I_{2B}$

$V_1 = Z_{11}\, I_1 + Z_{12}\, I_2$

$V_2 = Z_{21}\, I_1 + Z_{22}\, I_2$

$V_1 = V_{1A} + V_{1B}$

$V_2 = V_{2A} + V_{2B}$

$I_1 = I_{1A} = I_{1B}$

$I_2 = I_{2A} = I_{2B}$

$V_1 = V_{1A} + V_{1B} = (Z_{11A} + Z_{11B})\, I_1 + (Z_{12A} + Z_{12B}) I_2$

$V_2 = V_{2A} + V_{2B} = (Z_{21A} + Z_{21B})\, I_1 + (Z_{22A} + Z_{22B}) I_2$

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} (Z_{11A} + Z_{11B}) & (Z_{12A} + Z_{12B}) \\ (Z_{21A} + Z_{21B}) & (Z_{22A} + Z_{22B}) \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

$[Z] = [Z]_A + [Z]_B$

Note: For a series connection of two networks, the Z-parameter matrices are added.

ii) Parallel interconnection:

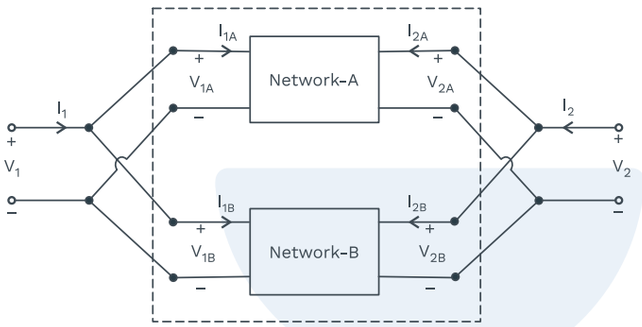


Fig. 4.23 Representation of parallel interconnection.

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

$[Y] = [Y]_A + [Y]_B$

For A-network

$I_{1A} = Y_{11A} V_{1A} + Y_{12A} V_{2A}$

$I_{2A} = Y_{21A} V_{1A} + Y_{22A} V_{2A}$

For B-network

$I_{1B} = Y_{11B} V_{1B} + Y_{12B} V_{2B}$

$I_{2B} = Y_{21B} V_{1B} + Y_{22B} V_{2B}$

As, $I_1 = I_{1A} + I_{1B}$

$I_2 = I_{2A} + I_{2B}$

$V_1 = V_{1A} = V_{1B}$

$V_2 = V_{2A} = V_{2B}$

$I_1 = (Y_{11A} + Y_{11B}) V_1 + (Y_{12A} + Y_{12B}) V_2$

$I_2 = (Y_{21A} + Y_{21B}) V_1 + (Y_{22A} + Y_{22B}) V_2$

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} Y_{11A} + Y_{11B} & Y_{12A} + Y_{12B} \\ Y_{21A} + Y_{21B} & Y_{22A} + Y_{22B} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

$[Y] = [Y]_A + [Y]_B$

Grey Matter Alert!

The two port network analysis is important for designing the transistors, transformers, transmission lines, amplifier circuits modelling etc.

Note: For a parallel connection of two networks, the y-parameters matrices are added.

iii) Series - Parallel interconnection:

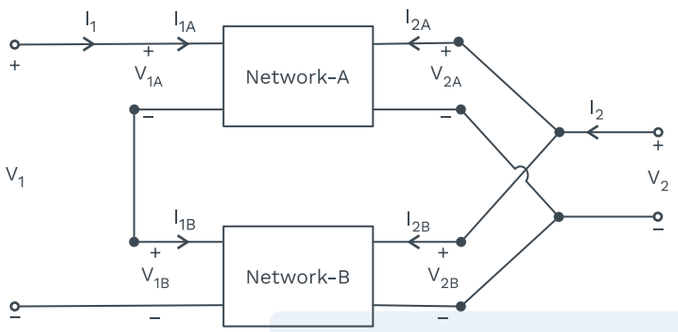


Fig. 4.24 Representation Series- parallel interconnection

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$

$$[h] = [h]_A + [h]_B$$

For A-network

$$V_{1A} = h_{11A}\, I_{1A} + h_{12A}\, V_{2A}$$

$$I_{2A} = h_{21A}\, I_{1A} + h_{22A}\, V_{2A}$$

For B- network

$$V_{1B} = h_{11B}\, I_{1B} + h_{12B}\, V_{2B}$$

$$I_{2B} = h_{21B}\, I_{1B} + h_{22B}\, V_{2B}$$

And

$$V_1 = V_{1A} + V_{1B}$$

$$I_1 = I_{1A} = I_{1B}$$

And

$$V_2 = V_{2A} = V_{2B}$$

$$I_2 = I_{2A} + I_{2B}$$

$$V_1 = (h_{11A} + h_{11B})I_1 + (h_{12A} + h_{12B})V_2$$

$$I_2 = (h_{21A} + h_{21B})I_1 + (h_{22A} + h_{22B})V_2$$

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11A} + h_{11B} & h_{12A} + h_{12B} \\ h_{21A} + h_{21B} & h_{22A} + h_{22B} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$

$$[h] = [h]_A + [h]_B$$

Note: For a Series-parallel connection of two networks, the h-parameters matrices are added.

iv) Parallel- series interconnection:

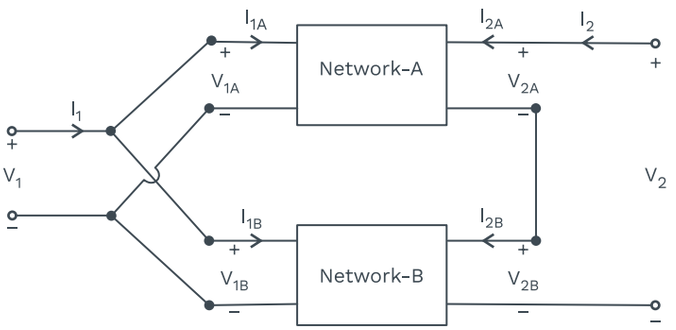


Fig. 4.25 Representation of parallel - series interconnection

$$V_1 = V_{1A} = V_{1B}$$
$$I_1 = I_{1A} + I_{1B}$$
$$V_2 = V_{2A} + V_{2B}$$
$$I_2 = I_{2A} = I_{2B}$$
$$\begin{bmatrix} I_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ I_2 \end{bmatrix}$$
$$\begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} = \begin{bmatrix} g_{11A} + g_{11B} & g_{12A} + g_{12B} \\ g_{21A} + g_{21B} & g_{22A} + g_{22B} \end{bmatrix}$$
$$[g] = [g]_A + [g]_B$$

Note: For a parallel-series connection of two networks, the g-parameters matrices are added.

v) Cascade connection:

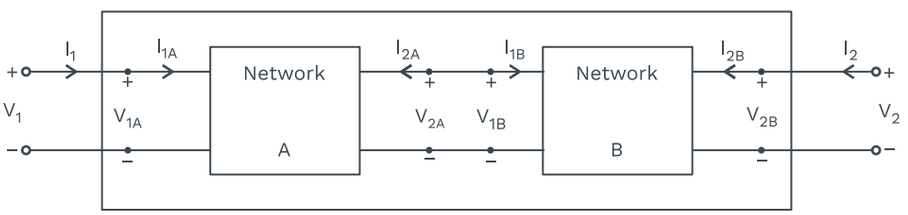


Fig. 4.26 Representation of cascade connection

$$[ABCD] = [ABCD]_A\,[ABCD]_B$$

* $\quad V_{1A} = A_A V_{2A} - B_A I_{2A}$ $\qquad \{V_1 = V_{1A}\}$
$\quad I_{1A} = C_A V_{2A} - D_A I_{2A}$ $\qquad \{I_1 = I_{1A}\}$

$$V_{1B} = A_B V_{2B} - B_B I_{2B}$$
$$I_{1B} = C_B V_{2B} - D_B I_{2B}$$

$$\left\{\begin{matrix} V_{2A} = V_{1B} \\ I_{2A} = -I_{1B} \end{matrix}\right\}$$

$$\left\{\begin{matrix} V_2 = V_{2B} \\ I_2 = I_{2B} \end{matrix}\right\}$$

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

$$\begin{bmatrix} V_{1A} \\ I_{1A} \end{bmatrix} = \begin{bmatrix} A_A & B_A \\ C_A & D_A \end{bmatrix} \begin{bmatrix} V_{2A} \\ -I_{2A} \end{bmatrix}$$

$$\begin{bmatrix} V_{1B} \\ I_{1B} \end{bmatrix} = \begin{bmatrix} A_B & B_B \\ C_B & D_B \end{bmatrix} \begin{bmatrix} V_{2B} \\ -I_{2B} \end{bmatrix}$$

$$\begin{bmatrix} V_{2A} \\ -I_{2A} \end{bmatrix} = \begin{bmatrix} A_B & B_B \\ C_B & D_B \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A_A & B_A \\ C_A & D_A \end{bmatrix} \begin{bmatrix} A_B & B_B \\ C_B & D_B \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A_A & B_A \\ C_A & D_A \end{bmatrix} \begin{bmatrix} A_B & B_B \\ C_B & D_B \end{bmatrix}$$

Note: For a cascade connection of two networks, the transmission line (T) parameters matrices are multiplied.

Chapter Summary



⇒ **Two port parameters:**

1) Impedance or Z-parameters:

$V_1 = Z_{11}I_1 + Z_{12}I_2$

$V_2 = Z_{21}I_1 + Z_{22}I_2$

- Z-parameters are also known as "open circuit parameters".

2) Y-parameters or Admittance parameters:

$I_1 = Y_{11}V_1 + Y_{12}V_2$

$I_2 = Y_{21}V_1 + Y_{22}V_2$

- Y-parameters are also known as "short circuit parameters".

3) h-parameters or Hybrid parameters:

$V_1 = h_{11}I_1 + h_{12}V_2$

$I_2 = h_{21}I_1 + h_{22}V_2$

- h-parameters are also known as “Hybrid parameters”.

4) g-parameters:

$I_1 = g_{11}V_1 + g_{12}I_2$

$V_2 = g_{21}V_1 + g_{22}I_2$

- g-parameters are also known as "inverse hybrid parameters".

5) ABCD-parameters:

$V_1 = AV_2 - BI_2$

$I_1 = CV_2 - DI_2$

- ABCD-parameters are also known as "transmission parameters".

⇒ Condition for a network to be reciprocal and symmetrical:

Reciprocal	Symmetrical
$Z_{12} = Z_{21}$	$Z_{11} = Z_{22}$
$Y_{12} = Y_{21}$	$Y_{11} = Y_{22}$
$\begin{vmatrix} A & B \\ C & D \end{vmatrix} = 1$	$A = D$
$h_{12} = -h_{21}$	$\begin{vmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{vmatrix} = 1$
$g_{12} = -g_{21}$	$\begin{vmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{vmatrix} = 1$

Table 4.2

⇒ **Interconnection of two port networks:**

1) Cascaded network:

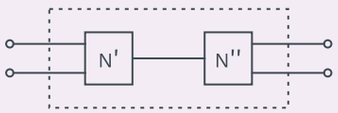


Fig. 4.27

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A' & B' \\ C' & D' \end{bmatrix} \begin{bmatrix} A'' & B'' \\ C'' & D'' \end{bmatrix}$$

- For a cascade connection of two networks, the transmission parameter matrices N′ and N″ are multiplied.

2) Series network:

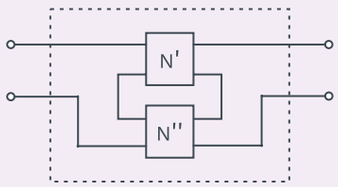


Fig. 4.28

$$\begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix} = \begin{bmatrix} z'_{11} + z''_{11} & z'_{12} + z''_{12} \\ z'_{21} + z''_{21} & z'_{22} + z''_{22} \end{bmatrix}$$

- For a series connection of two networks, the Z-parameter matrices N′ and N″ are added.

3) Parallel network:

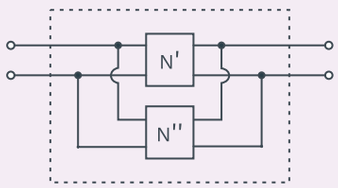


Fig. 4.29

$$\begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} = \begin{bmatrix} y'_{11} + y''_{11} & y'_{12} + y''_{12} \\ y'_{21} + y''_{21} & y'_{22} + y''_{22} \end{bmatrix}$$

- For a parallel connection of two networks, the y-parameter matrices N′ and N″ are added.

4) Series-parallel network:

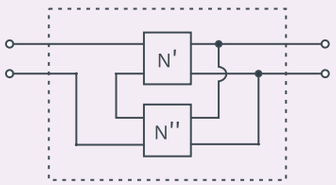


Fig. 4.30

$$\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} h'_{11} + h''_{11} & h'_{12} + h''_{12} \\ h'_{21} + h''_{21} & h'_{22} + h''_{22} \end{bmatrix}$$

- For a series-parallel connection of two networks, the h-parameter matrices N′ and N″ are added.

5) Parallel-series networks:

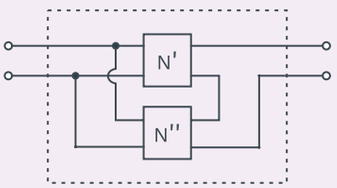


Fig. 4.31

$$\begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} = \begin{bmatrix} g'_{11} + g''_{11} & g'_{12} + g''_{12} \\ g'_{21} + g''_{21} & g'_{22} + g''_{22} \end{bmatrix}$$

- For a parallel-series connection of two networks, the g-parameter matrices N′ and N″ are added.

⇒ **Lattice network:**

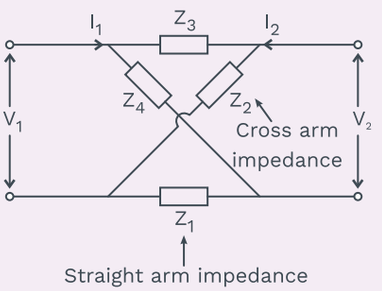


Fig. 4.32

1) A lattice network is considered to be symmetrical if the two straight arm impedances are equal.

i.e. $Z_1 = Z_3$

Therefore, $Z_{11} = Z_{22}$

2) A lattice network is considered to be reciprocal if two cross arm impedances are equal.

i.e. $Z_2 = Z_4$

Therefore, $Z_{12} = Z_{21}$

PRACTICE QUESTIONS

Q1 **Obtain the y-parameters for the circuit shown in the figure.**

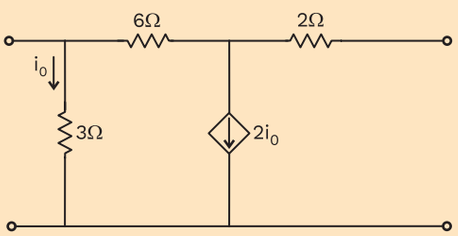


Fig. 4.31

Sol: $y_{11} = 0.625$ S, $y_{12} = -0.125$ S, $y_{21} = 0.375$ S, $y_{22} = 0.125$ S.

Q2 **Determine the h-parameters for the circuit shown in the figure.**

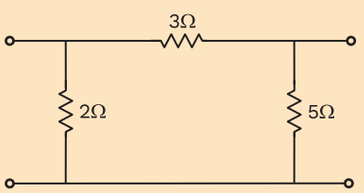


Fig. 4.32

Sol: $h_{11} = 1.2\ \Omega$, $h_{12} = 0.4$, $h_{21} = -0.4$, $h_{22} = 0.4$ S.

Q3 **Find the impedance at the input port of the circuit shown in the figure.**

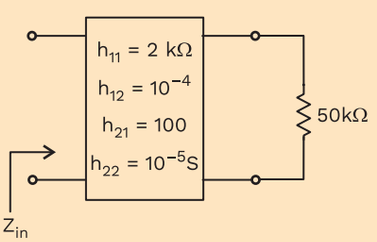


Fig. 4.33

Sol: $Z_{in} = 1667\ \Omega$.

Q4 **For the two-port network shown in the figure, find the transmission parameters.**

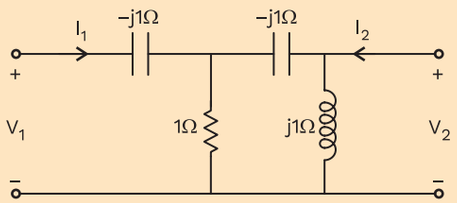


Fig. 4.34

Sol: $[T] = \begin{bmatrix} -1 & (-1-j2)\Omega \\ -jS & (1-j) \end{bmatrix}$

Q5 **Find [z] and [g] of a two port networks if**

$$\mathbf{[T]} = \begin{bmatrix} \mathbf{10} & \mathbf{1.5\Omega} \\ \mathbf{2S} & \mathbf{4} \end{bmatrix}$$

Sol: $[z] = \begin{bmatrix} 5 & 18.5 \\ 0.5 & 2 \end{bmatrix}\Omega,\ [g] = \begin{bmatrix} 0.2S & -3.7 \\ 0.1 & 0.15\Omega \end{bmatrix}$

Q6 **Determine [y] and [T] of a two port network whose z-parameters are**

$$\mathbf{[z]} = \begin{bmatrix} \mathbf{6} & \mathbf{4} \\ \mathbf{4} & \mathbf{6} \end{bmatrix}\Omega$$

Sol: $[y] = \begin{bmatrix} 0.3 & -0.2 \\ -0.2 & 0.3 \end{bmatrix}S,\ [T] = \begin{bmatrix} 1.5 & 5\Omega \\ 0.25S & 1.5 \end{bmatrix}$

Q7 **Find the z-parameters of the op amp circuit in the figure. Show that the circuit has no y-parameters.**

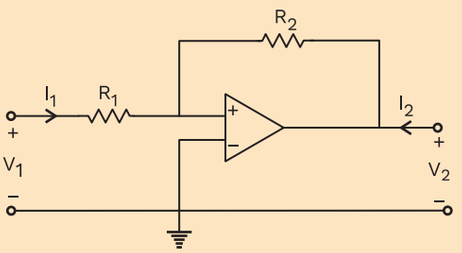


Fig. 4.35

Sol: $[z] = \begin{bmatrix} R_1 & 0 \\ -R_2 & 0 \end{bmatrix}$. Since $[z]^{-1}$ does not exist, [y] does not exist.